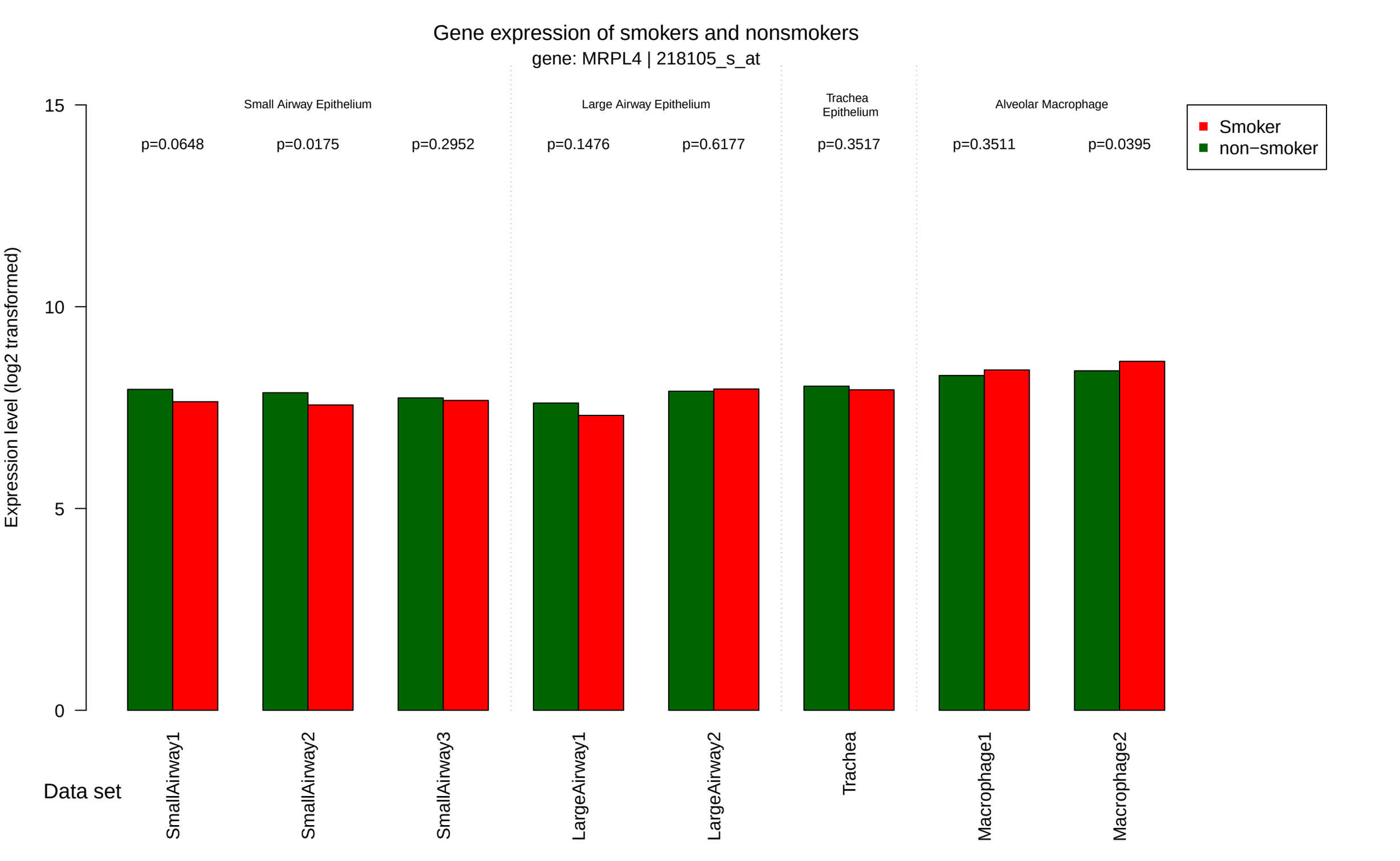
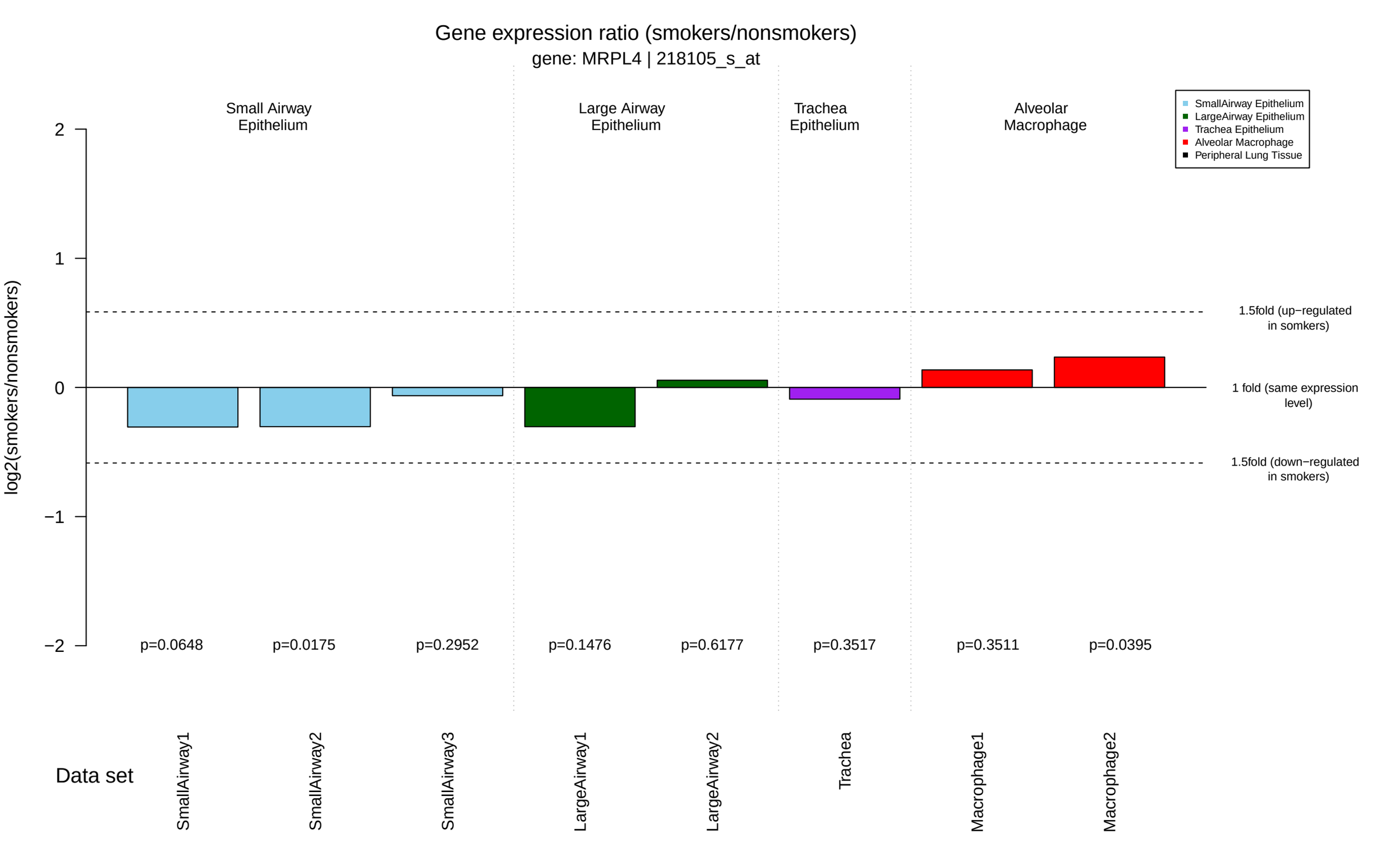
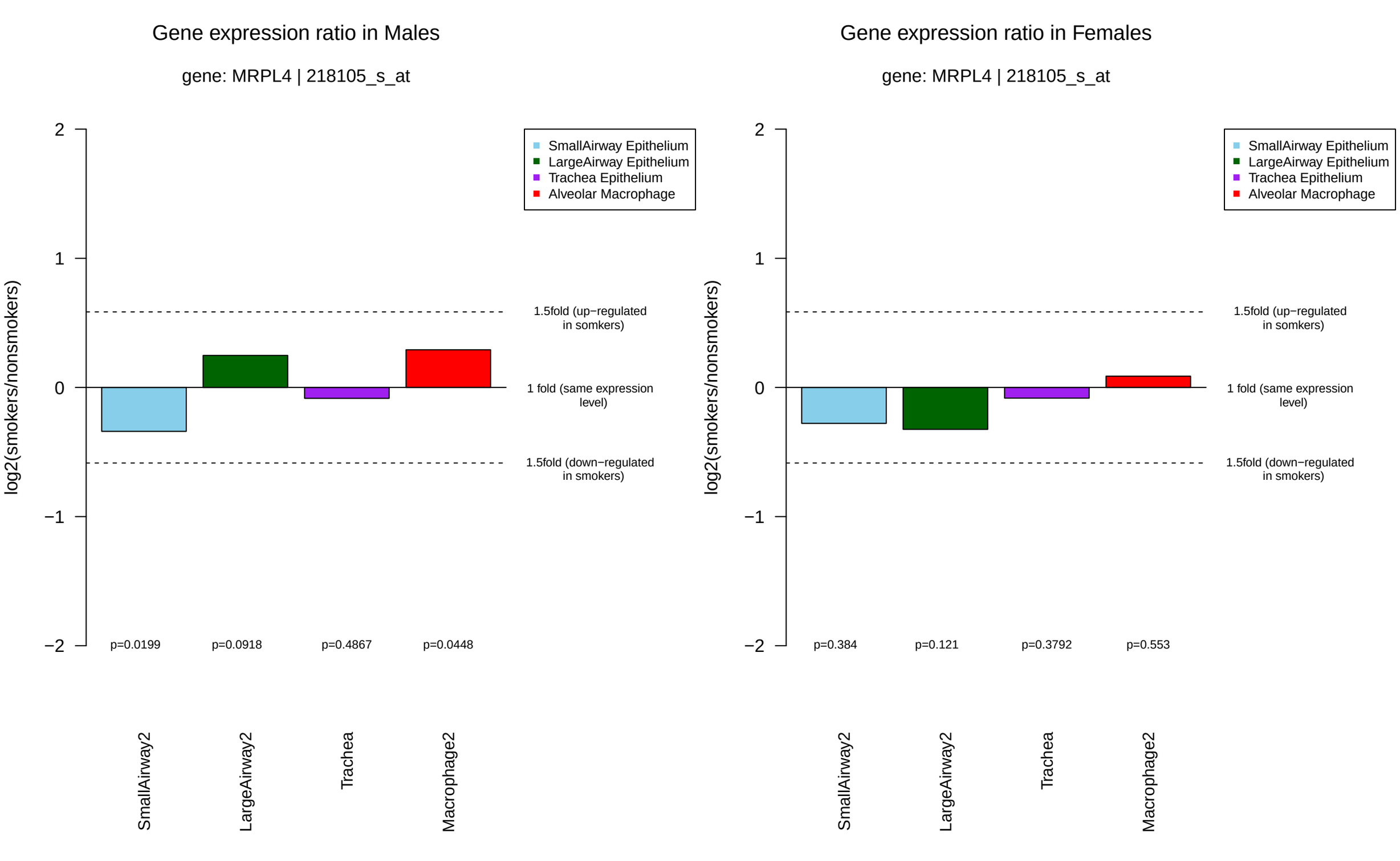
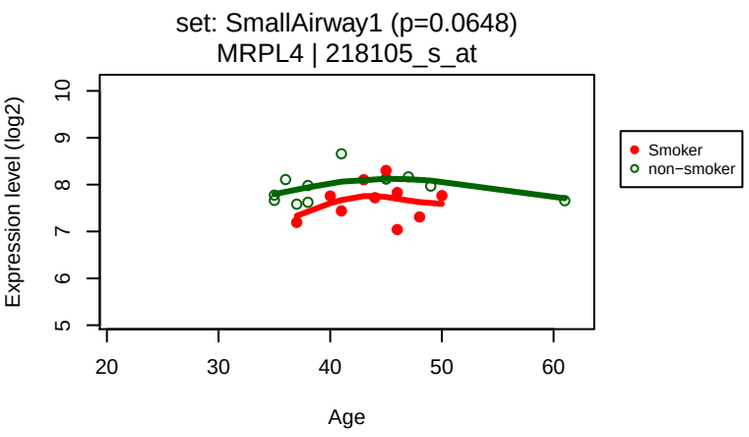
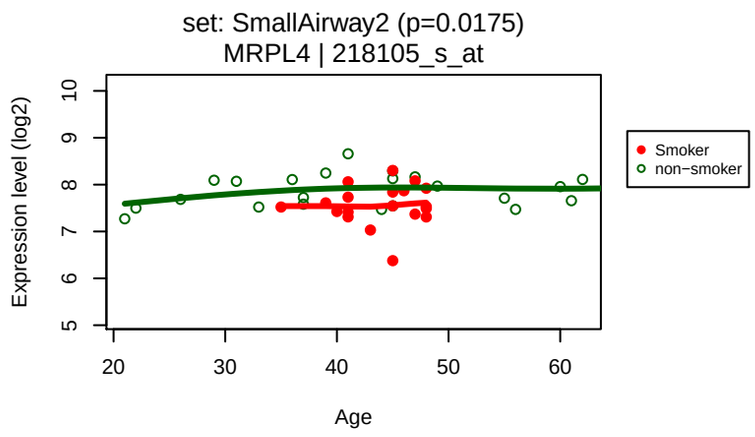
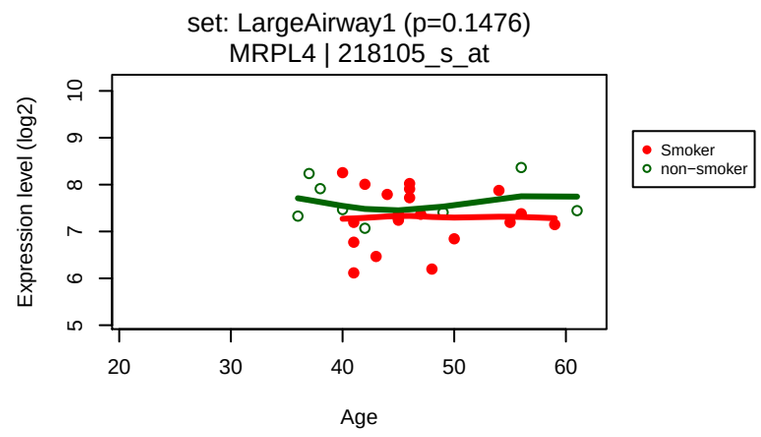

Gene expression of smokers and nonsmokers
gene: MRPL4 | 218105_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0648
p=0.0175
p=0.2952
p=0.1476
p=0.6177
p=0.3517
p=0.3511
p=0.0395
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

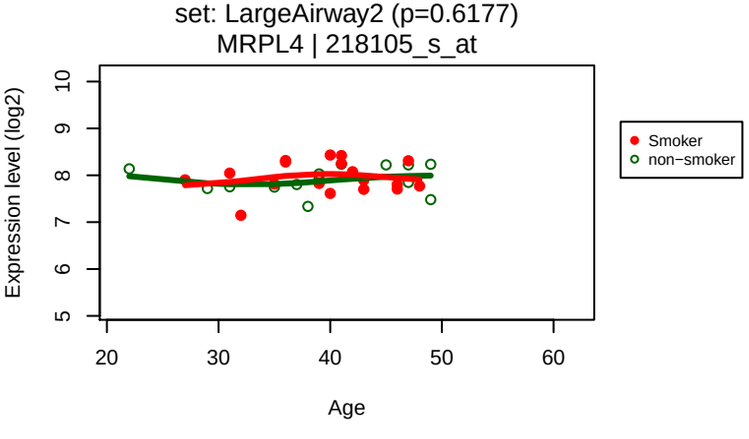

Gene expression ratio (smokers/nonsmokers)
gene: MRPL4 | 218105_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0648
p=0.0175
p=0.2952
p=0.1476
p=0.6177
p=0.3517
p=0.3511
p=0.0395
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

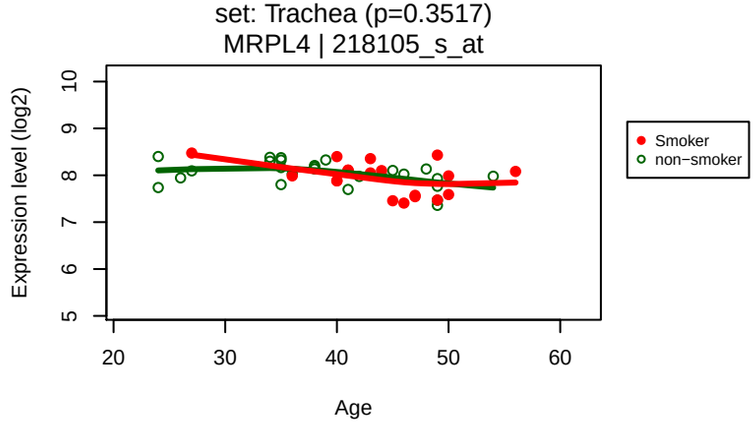

Gene expression ratio in Males
gene: MRPL4 | 218105_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0199
p=0.0918
p=0.4867
p=0.0448
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL4 | 218105_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.384
p=0.121
p=0.3792
p=0.553
SmallAirway2
LargeAirway2
Trachea
Macrophage2

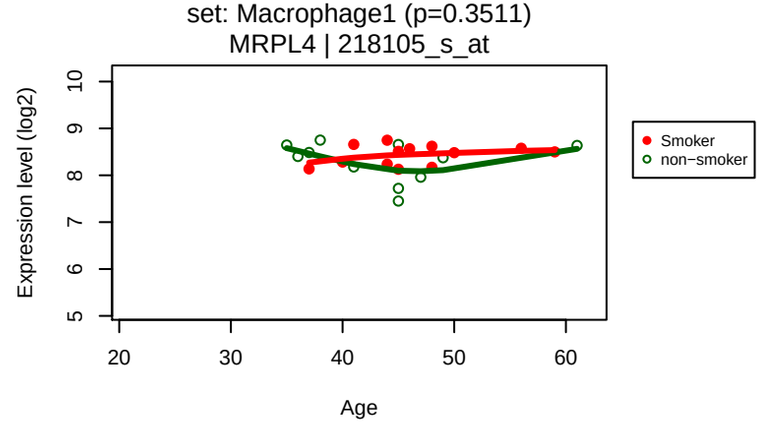

set: SmallAirway1 (p=0.0648)
MRPL4 | 218105_s_at
Expression level (log2)
Age
Smoker
non−smoker

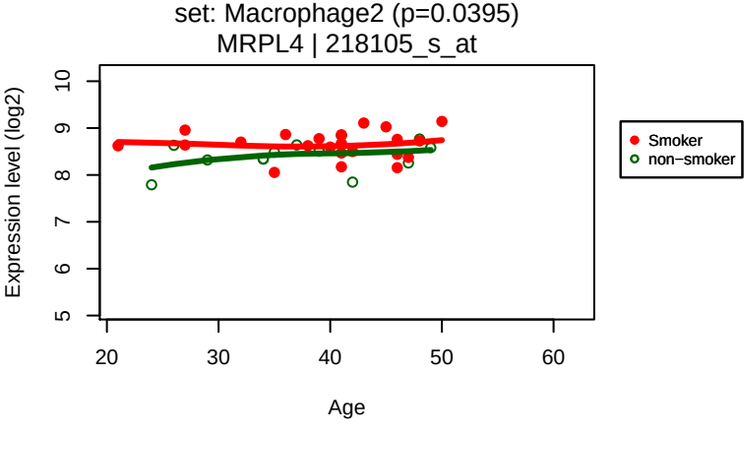

set: SmallAirway2 (p=0.0175)
MRPL4 | 218105_s_at
Expression level (log2)
Age
Smoker
non−smoker

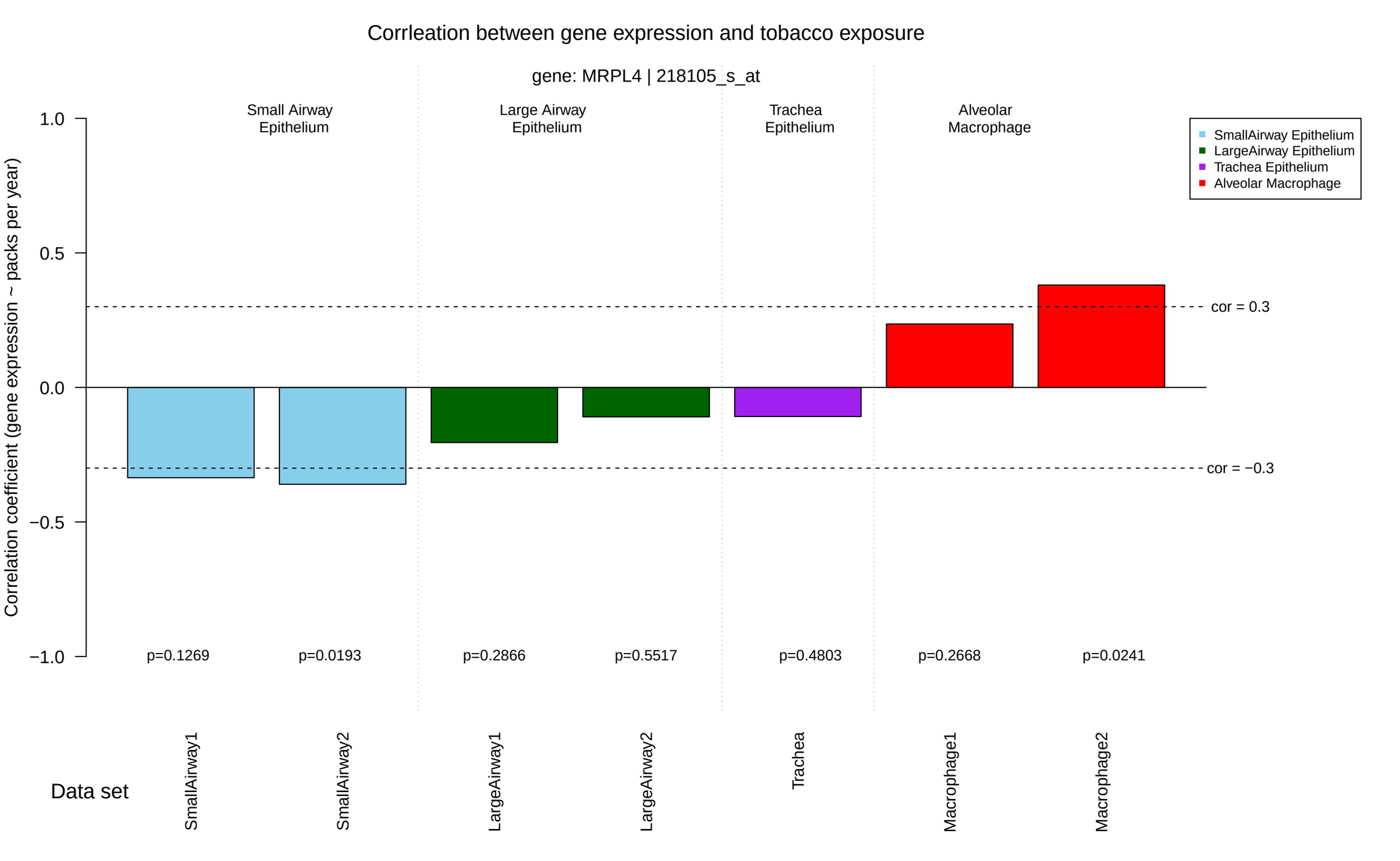

set: LargeAirway1 (p=0.1476)
MRPL4 | 218105_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6177)
MRPL4 | 218105_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3517)
MRPL4 | 218105_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3511)
MRPL4 | 218105_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0395)
MRPL4 | 218105_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL4 | 218105_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1269
p=0.0193
p=0.2866
p=0.5517
p=0.4803
p=0.2668
p=0.0241
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2